EXERCISE 11.1

1. Suppose X is the number of tails occurred when three fair coins are tossed once simultaneously. Find the values of the random variable X and number of points in its inverse images.
2. An urn contains 5 mangoes and 4 apples. Three fruits are taken at random. If the number of apples taken is a random variable, then find the values of the random variable and number of points in its inverse images.
3. Two balls are chosen randomly from an urn containing 6 red and 8 black balls. Suppose that we win ₹ 15 for each red ball selected and we lose ₹ 10 for each black ball selected. If X denotes the winning amount, find the values of X and number of points in its inverse images.
4. A six sided die is marked '2' on one face, '3' on two of its faces, and '4' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find the values of the random variable and number of points in its inverse images.

11.3 Types of Random Variable

In this chapter we shall restrict our study to two types of random variables, one is a random variable assuming at most a countable number of values and another is a random variable assuming the values continuously. That is

(i) Discrete Random variable (for counting the quantity)

(ii) Continuous Random variable (for measuring the quantity)

11.3.1 Discrete random variables

In this section we discuss

(i) Discrete random variables

(ii) Probability mass function

(iii) Cumulative distribution function.

(iv) Obtaining cumulative distribution function from probability mass function.

(v) Obtaining probability mass function from cumulative distribution function.

If the range set of the random variables is discrete set of numbers then the inverse image of random variable is either finite or countably infinite. Such a random variable is called discrete random variable. A random variable defined on a discrete sample space is discrete.

Definition 11.2 (Discrete Random Variable)

A random variable X is defined on a sample space S into the real numbers $\mathbb{R}$ is called discrete random variable if the range of X is countable, that is, it can assume only a finite or countably infinite number of values, where every value in the set S has positive probability with total one.

Remark

It is also possible to define a discrete random variable on continuous sample space. For instance,

(i) for a continuous sample space $S = [0,1]$, the random variable defined by $X(\omega) = 10$ for all $\omega \in S$, is a discrete random variable.

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 184 16-12-2022 19:18:35